Compliance Test Report
Utility Flare Performance Test

Wexford County Landfill
Manton, Michigan

May 22, 2026

Prepared for:
Wexford County Landfill
990 US Highway 131 North
Manton, Michigan 49724

Prepared by:
Environmental Information Logistics, LLC

May 22, 2026

Ms. Lindsey Wells
Michigan Department of Environment, Great Lakes, and Energy
Air Quality Division, Cadillac District Office
120 West Chapin Street
Cadillac, MI 49601

Re: Utility Flare Performance Test Report Revision
Wexford County Landfill (MI-ROP-N3862-2022)
SRN: N3862

Dear Ms. Wells:

On behalf of Wexford County Landfill (Wexford), Environmental Information Logistics, LLC. (EIL) is pleased to provide you the revised compliance performance evaluation test results for the utility flare located at Wexford County Landfill in Manton, Michigan.

The purpose of the test program was to demonstrate that the utility flare meets the performance requirements of, and is thus in compliance, with §63.11(b) of the Landfill NESHAP and 40 CFR §60.18 as referenced in §62.16714(c)(1) of the Federal Plan.

The utility flare was tested on February 27, 2026, at maximum expected operating conditions and in accordance with the test plan dated January 22, 2026. The test results demonstrate that the utility flare meets the performance criteria of §63.11(b) and §60.18. These test results were originally submitted on April 24, 2026. A calculation error was discovered, resulting in this revised report.

If you have any questions, or require additional information, please contact me by phone at (989) 415-3741, or bkotrba@eilllc.com.

Sincerely,

Benjamin Kotrba
Environmental Scientist

cc: Chris Gee – GFL Environmental
Wexford County Landfill (Operating Record)

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CONTENTS

EXECUTIVE SUMMARY

Wexford County Landfill (Wexford) retained Environmental Information Logistics, LLC (EIL) to conduct a performance evaluation of the utility (open) flare located at Wexford County Landfill in Manton, Michigan. The utility flare is the primary control device to control landfill gas emissions from Wexford County Landfill.

The purpose of the test program was to demonstrate that the utility flare meets the performance requirements of, and is thus in compliance, with §63.11(b) of the Landfill NESHAP and 40 CFR §60.18 as referenced in §62.16714(c)(1) of the Federal Plan.

EIL conducted the fieldwork on February 27, 2026, and in accordance with the Test Plan, dated January 22, 2026. Mr. Ben Kotrba and Mr. Tyler Smith conducted the tests. Mr. Chris Gee with Wexford provided on-site coordination of the tests with landfill operations. Ms. Lindsey Wells with Michigan Department of Environment, Great Lakes, and Energy (EGLE) reviewed and approved the test plan.

The results of the performance evaluations were:

Parameter	Applicable Requirement	Average Test Result
Flare Exhaust Smoke Emissions (Visual Emissions in a 2-hour Period)	<5 minutes over 2 hours [1]	0 minutes, 3 seconds
Flare Inlet Gas Net Heating Value (MJ/scm)	>7.45 [2]	16.01
Flare Exhaust Gas Exit Velocity (Feet per second)	<60 [3]	56.31
Maximum Permitted Velocity (V_{max}, feet per second)	<61.98[4]	56.31

MJ: megajoules
scm: standard cubic meter

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[1] 40 CFR 63.11(b)(4), 60.18(c)(1)
[2] 40 CFR 63.11(b)(6)(ii), 60.18(c)(3)(ii)
[3] 40 CFR 63.11(b)(7)(i), 60.18(c)(4)(i)
[4] 40 CFR 63.11(b)(7)(iii), 60.18(c)(4)(iii)

1.0 INTRODUCTION

Wexford County Landfill (Wexford) retained Environmental Information Logistics, LLC (EIL) to conduct a performance evaluation of the utility (open) flare located at Wexford in Manton, Michigan. The utility flare is the primary control device to control landfill gas emissions from Wexford County Landfill.

The purpose of the test program was to demonstrate that the utility flare meets the performance requirements of ,and is thus in compliance, with §63.11(b) of the Landfill NESHAP and 40 CFR §60.18 as referenced in §62.16714(c)(1) of the Federal Plan.

EIL conducted the test program with methodologies outlined in 40 CFR 63.11(b) and 60.18, except that United States Environmental Protection Agency (USEPA) Method 3C, *"Determination of Carbon Dioxide, Methane, Nitrogen, and Oxygen from Stationary Sources,"* was employed for net heating value determination in lieu of Method 18 and ASTM D1946. Method 3C is the applicable method for utility flares at landfills, in accordance with 40 CFR §63.1959(e) and 40 CFR §62.16718(d).

EIL conducted the fieldwork on February 27, 2026, and in accordance with the Test Plan, dated January 22, 2026. Mr. Ben Kotrba and Mr. Tyler Smith conducted the tests. Mr. Chris Gee with Wexford provided on-site coordination of the tests with landfill operations. Ms. Lindsey Wells with Michigan Department of Environment, Great Lakes, and Energy (EGLE) reviewed and approved the test plan.

The name, address, and telephone number of the primary contact for further information about the tests and this test report is:

Name and Title	**Company**	**Telephone**
Mr. Ben Kotrba Environmental Scientist	Environmental Information Logistics, LLC 707 South Chilson Street Bay City, MI 48706	(989) 415-3741

The name, address, and telephone number of the primary contact for further information about the flare and associated operations is:

Name and Title	**Company**	**Telephone**
Mr. Chris Gee General Manager	GFL Environmental 1307 Higgins Drive Cheboygan, Michigan 49721	(231) 597-8553

2.0 SUMMARY OF RESULTS

On February 27, 2026, the utility flare operated at an average inlet volumetric flow rate of approximately 1,179 scfm as averaged from the process flow meter data.

The average test results were:

1) visible emissions: 0 minutes, 3 seconds (accumulated, total),
2) average net heating value of the gas being combusted: 16.01 megajoules per standard cubic meter (MJ/scm), and
3) average exhaust gas exit velocity: 56.31 feet per second (fps).

The performance criteria are less than 5 minutes visible emissions in a 2-hour period, a net heating value of greater than 7.45 MJ/scm, and an exit velocity less than 60 fps (or less than the maximum permitted velocity (V_{max}), calculated to be 61.98 fps).

The test results demonstrate that that utility flare meets the performance requirements of §63.11(b), §60.18, and thus also satisfies the requirements of 63.1959(b)(2)(iii)(B) and 62.16714(c)(2), at the test flow rate.

3.0 SOURCE DESCRIPTION

Wexford is an active municipal solid waste (MSW) landfill. Anaerobic bacteria decompose the emplaced waste. The primary by-products of decomposition are methane (~40-50%, typical) and carbon dioxide (~35-45%, typical), with the remainder balance gases nitrogen, oxygen, and trace amounts of non-methane organic compounds.

Wexford employs a gas collection and control system to meet the requirements of the Landfill NESHAP and the Federal Plan. Gas collection wells are installed in a grid pattern about the landfill. The wells are connected to a common header system. A blower produces a vacuum on the well field. Collected gas is routed to treatment systems for beneficial use. The utility flare serves as the primary control for the landfill gas collection system.

The utility flare is designed to meet the requirements of 63.1959(b)(2)(iii)(B) and 62.16714(c)(2) at a flow rate up to 1,600 scfm. The utility flare was tested at a flow rate of approximately 1,179 scfm as measured by the installed process flow meter.

The landfill gas flow rate is variable and depends on gas production in the landfill. The composition of the landfill gas varies, but the average Method 3C values obtained on March 16, 2026, may be considered 'typical' for the gas quality directed to the flare: methane, 48.10%; carbon dioxide, 31.87%; oxygen, 4.34%; and nitrogen, 24.50%.

The utility flare is equipped with a thermocouple to monitor for the presence of a flame, and an automatic shutdown software routine that activates if the presence of flame cannot be verified by the sensor.

4.0 SAMPLE AND ANALYTICAL PROCEDURES

EIL conducted measurements in accordance with USEPA Reference Test Methods, as presented in 40 CFR 63 and 62, Appendix A. The sample collection and analytical methods used in the test program are listed in the table below. Figure 1 depicts the sample site.

Sample Method	Parameter	Analysis
USEPA Method ALT-073	Stack Gas Velocity and Volumetric Flow Rate	Measured flow rate from calibrated mass flow meter
USEPA Method 3C	Carbon Dioxide, Methane, Nitrogen, Oxygen, and moisture fraction	Gas Chromatography / Thermal Conductivity Detector (GC/TCD)
USEPA Method 22	Visible Emissions	Field Observation

4.1 Stack Gas Velocity and Volumetric Flow Rate (USEPA Method ALT-073)

The exit velocity from the flare stack was determined based on landfill gas flow rates measured by the calibrated mass flow meter, utilizing USEPA Method ALT-073. This alternative method of determining the flow described in 40 CFR 60.18(f)(4) received a blanket approval from the US EPA Office of Air Quality Planning and Standards in a letter to SCS Engineers in 2009. A copy of the letter was provided with the alternative method request submitted to EGLE on February 20, 2026.

Flow rate data were recorded by the flare's digital data logger and downloaded for the evaluation. Flow meter calibration documentation and the flare operational data are included in Appendix A.

The flow rate to the flare was also recorded before and after each Method 3C gas sample run. Those measurements are shown in field data logs included as Appendix B. However, these values were not used in the subsequent calculations because the flow meter data from the data logger were deemed valid.

The average flow rates measured during each sample run by the calibrated flow meter, in standard units of temperature and pressure (see Appendix B), were converted from cubic feet per minute (ft^3/min) to cubic feet per second (ft^3/s). The flow rates (ft^3/s) were divided by the unobstructed cross-sectional area of the flare tip (0.3491 ft^2) to determine the exit velocity (ft/s) for each sample run.

4.2 Determination of Carbon Dioxide, Methane, Nitrogen, and Oxygen from Stationary Sources (Method 3C)

EIL used Method 3C to determine the composition of the landfill gas. EIL collected three, 30-minute (minimum), integrated tank samples of landfill gas from the utility flare inlet (downstream of the blower).

EIL submitted the samples to Enthalpy Analytical (EA), Richmond, Virginia for analysis. EA analyzed each tank for carbon dioxide (CO_2), methane (CH_4), nitrogen (N_2), and oxygen (O_2) concentration via Method 3C. Figure 3 depicts the Method 3C sample train.

EA followed the analytical procedures of Method 3C by using a gas chromatograph (GC), with appropriate separation column for the expected parameters, equipped with a thermal conductivity detector (TCD). The EA laboratory analytical report is presented in Appendix B.

EIL used the Method 3C analytical results to calculate stack gas molecular weight (for use in stack gas velocity calculation), and to calculate the maximum permitted velocity, V_{max}, per §63.11(b)(7)(i), 60.18(c)(4)(i). The reported net heating value is the arithmetic average of three valid test runs.

EIL calculated the dry molecular weight of the stack gas based on the primary constituents of methane, carbon dioxide, nitrogen, oxygen, and hydrogen (other compounds present have a negligible relative concentration). The stack gas dry molecular weight is equal to the sum of stack gas constituent concentrations (%) multiplied by the corresponding molecular weight of that constituent.

4.3 Visual Determination of Fugitive Emissions from Material Sources and Smoke Emissions from Flares (Method 22)

EIL conducted a single, 120-minute, non-continuous observation of the utility flare exhaust for smoke emissions. EIL observed continuously for 15 to 20 minutes, then took a break for at least 5 – but no more than 10 minutes and then resumed observation in this pattern until a full 120-minute period of observation time had accrued. A copy of the Method 22 observation data is presented in Appendix A.

5.0 RESULTS AND DISCUSSION

On February 27, 2026, EIL observed an accumulated total of 0 minutes, 3 seconds of visible emissions from the utility flare exhaust. The limit for visible emissions is less than 5 minutes per 2-hour time period [63.11(b)(4), 60.18(c)(1)].

On February 27, 2026, the average net heating value of the gas being combusted was 16.01 MJ/scm. The requirement for net heating value is >7.45 MJ/scm [63.11(b)(6)(ii), 60.18(c)(3)(ii)].

On February 27, 2026, the average stack gas exit velocity, calculated from field data, was 56.31 fps. The limit is <60 fps [63.11(b)(7)(i), 60.18(c)(4)(i)], or less than the Maximum Permitted Velocity, V_{max}, calculated to be 84.96 fps [63.11(b)(7)(iii), 60.18(c)(4)(iii)].

The February 27, 2026 results demonstrate that the utility flare meets the performance requirements of, and is thus in compliance with, §63.11(b) of the Landfill NESHAP and 40 CFR §60.18 as referenced in §62.16714(c)(1) of the Federal Plan.

EIL quality assurance (QA) procedures included:

1) leak-check of the velocity measurement system (pitot tube through manometer), prior to each test,
2) leak-check of the Method 3C train, prior to each test, and,
3) verification of sufficient evacuation of each Method 3C canister prior to initiation of each sample collection.

Process flow data and the calibration certificate for the mass flow meter are provided in Appendix A. Raw field and computer-calculated data used in the determination of the utility flare average exit velocities and net heating values, visible emissions observation data are presented in Appendix B. The Method 3C laboratory analytical results and chain-of-custody forms are presented in Appendix C. Sample calculations are presented in Appendix D.

This report prepared by:

Benjamin Kotrba
Environmental Scientist
Environmental Information Logisitics

May 22, 2026

ENVIRONMENTAL INFORMATION LOGISTICS, LLC

TABLES

Table 1

Utility Flare Inlet Volumetric Flow Rate and Flare Exit Velocity
Wexford County Landfill
Manton, Michigan
February 27, 2026

Parameter	Test 1	Test 2	Test 3	Average
Inlet Volumetric Flow Rate (scfm) – Measured Field Data	1,180	1,178	1,178	1,179
Exit Tip Diameter (inches)	8	8	8	
Exit Tip Cross-Sectional Area (ft^2)	0.3491	0.3491	0.3491	
Allowable Exit Velocity (fps)[1]	**60**	**60**	**60**	**60**
Maximum Permitted Velocity, V_{max} (fps) [2]	**84.96**	**84.96**	**84.96**	**84.96**
Calculated Exit Velocity (fps)	**56.36**	**56.26**	**56.29**	**56.31**

[1] 40 CFR 63.11(b)(6)(i), 60.18(c)(3)(i)
[2] 40 CFR 63.11(b)(6)(i), 60.18(c)(3)(i)
scfm: standard cubic feet per minute
ft^2: square feet
fps: feet per second

ENVIRONMENTAL INFORMATION LOGISTICS, LLC

Table 2

Utility Flare Inlet Gas Net Heating Value
Wexford County Landfill
Manton, Michigan
February 27, 2026

Parameter	Test 1	Test 2	Test 3	Average
Flare Inlet Gas Methane Content (%)	47.80	47.50	49.00	48.10
Methane, Molecular Weight (lb/lb mole)	16	16	16	
Methane, Heating Value (kcal/g)[1]	11.9533	11.9533	11.9533	
Methane, Heating Value (kcal/g mole)	191.25	191.25	191.25	
Minimum Net Heating Value (MJ/scm)[2]	**7.45**	**7.45**	**7.45**	**7.45**
Flare Inlet Gas Net Heating Value (MJ/scm)	**15.91**	**15.81**	**16.31**	**16.01**

1 USEPA Office of Air Quality Planning and Standards' Control Cost Manual
2 40 CFR 63.11(b)(6)(ii), 60.18(c)(3)(ii)
ppm: parts per million
%: percent
lb/lb mole: pounds per pound-mole
kcal/g: kilocalories per gram
kcal/g mole: kilocalories per gram-mole
MJ/scm: megajoules per standard cubic meter

ENVIRONMENTAL INFORMATION LOGISTICS, LLC

FIGURES

ENVIRONMENTAL
INFORMATION
LOGISTICS, LLC

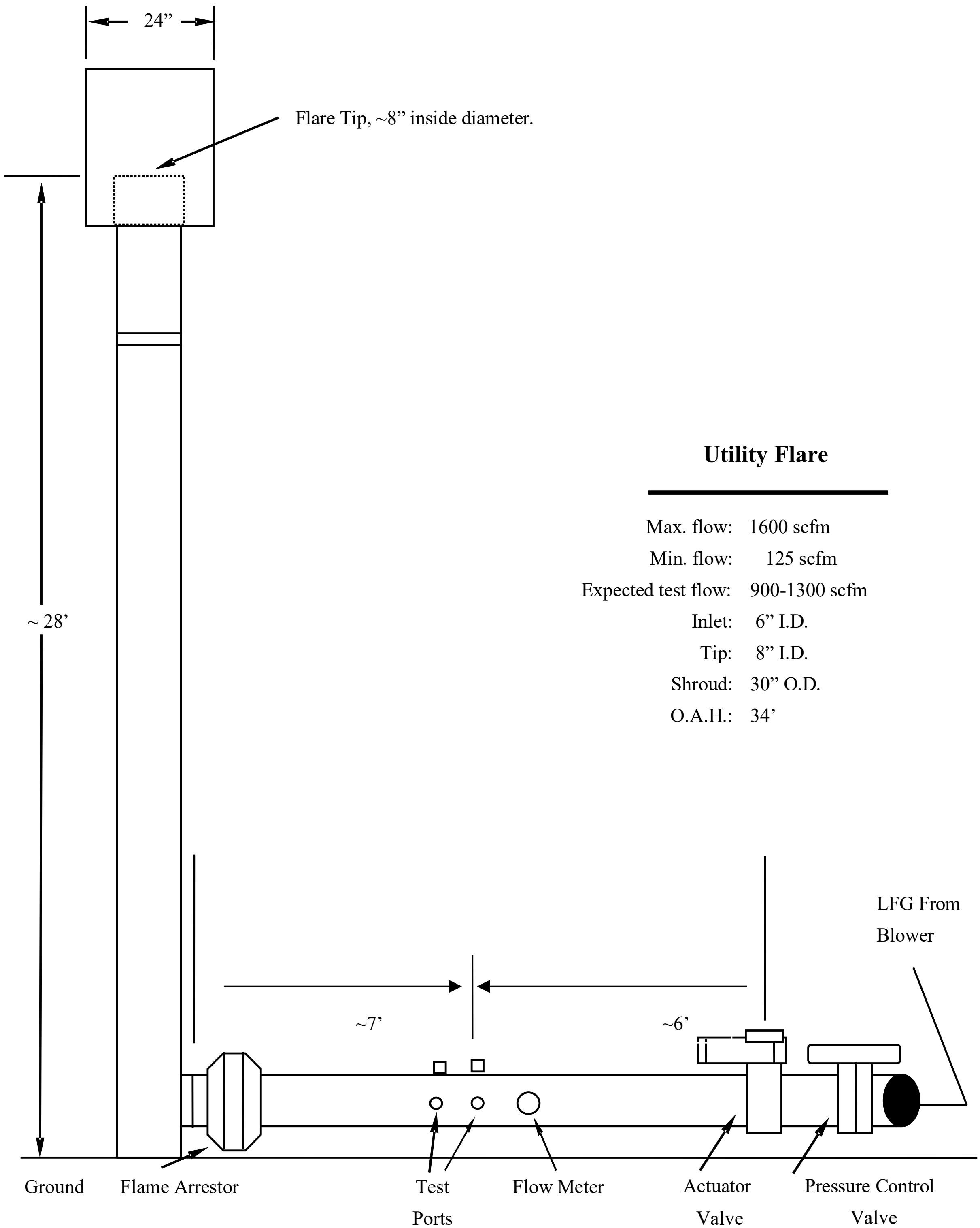

Figure 2
Utility flare duct and stack arrangement, approximate dimensions, and test locations, Wexford County Landfill in Manton, Michigan.

EIL LLC
2026

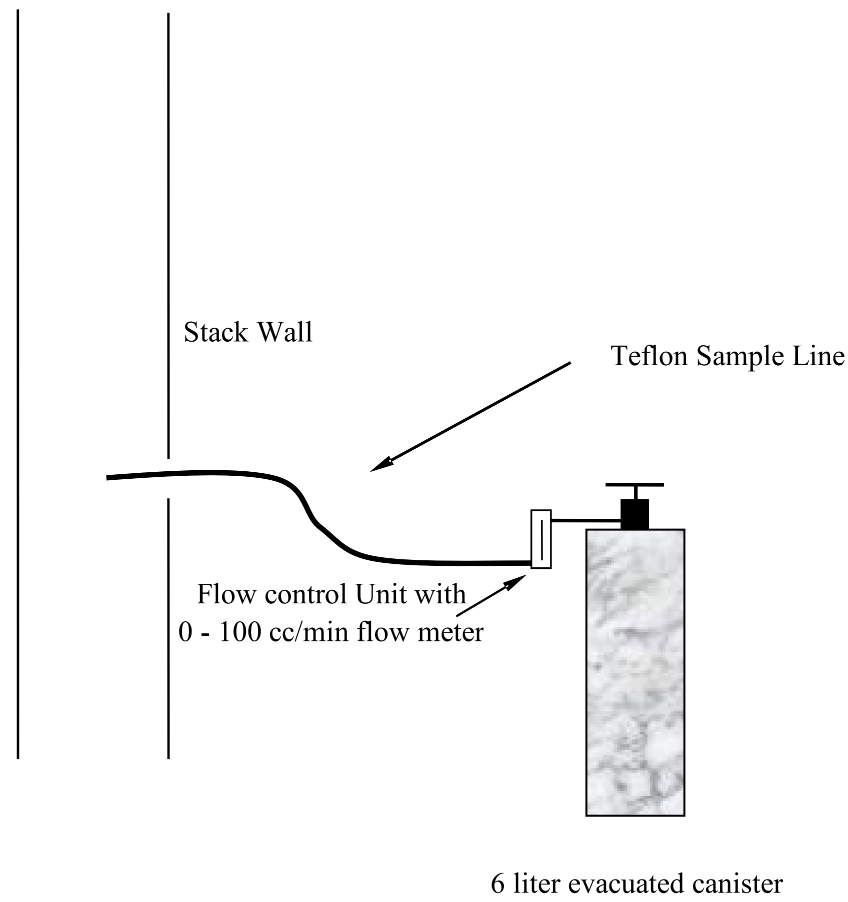

Figure 1
USEPA Method 3C sample train for the utility flare inlet duct at Wexford County Landfill in Manton, Michigan.

EIL LLC
2026

APPENDIX A

FLOW METER CALIBRATION AND FLARE OPERATIONAL DATA

THERMAL INSTRUMENT COMPANY, INC.

CALIBRATION CERTIFICATION

We certify that the calibration accuracies listed below are obtained on equipment, and with methods, that can be traced directly to the US National Institute of Standards and Technology.

FLOWRATE READOUT ACCURACY: ± 1% Full Scale

PRESSURE TESTED AT:

METER SERIAL NUMBER: **2025311**
MODEL NUMBER: **62-9TEF/9500PI**

The calibration listed above was performed under the following conditions:

0 - 2500 SCFM LANDFILL GAS MIX
40 - 200 DEG F
0 - 25 PSIG
6" SCH 10 LINE
6.357" LINE ID
LFG MIX: 49% CH_4, 38% CO_2, 1% O_2

Purchase Order: 26541-SPR

Standards:

Customer Address:
Landmarc IL
Kyle Gragg
3516 Ogden Rd.
Springfield IL 62711

Standard Conditions: 60Deg. F @ 1 ATM

Deviations:

CERTIFIED THAT THE SUPPLIES/SERVICES DETAILED HEREIN HAVE BEEN INSPECTED AND TESTED IN ACCORDANCE WITH THE CONDITIONS AND REQUIREMENTS OF THE CONTRACT OR PURCHASE ORDER, AND UNLESS OTHERWISE SPECIFIED AS AN AGREED DEVIATION FROM THE CONTRACT, CONFORM IN ALL RESPECTS TO THE SPECIFICATION(S) AND DRAWING(S) RELEVANT THERETO

Signature ______________________ Date **June 30, 2025**

UNITED STATES ENVIRONMENTAL PROTECTION AGENCY
RESEARCH TRIANGLE PARK, NC 27711

MAY 20 2009

OFFICE OF
AIR QUALITY PLANNING
AND STANDARDS

David H. Penoyer
SCS Engineers
4041 Park Oaks Boulevard, Suite 100
Tampa, FL 33610

Dear Mr. Penoyer:

In your May 1, 2009 letter, you requested permission to use an alternative procedure to measure the flow rate of gas exiting the candlestick flare at the Richland Creek Road Landfill in Buford, Georgia. The flare is subject to 40 CFR Part 60.18 for an open flare burning landfill gas. You would like to use a mass flowmeter in place of Method 2, 2A, 2C, or 2D to satisfy the requirements of 60.18(f)(4). Your flowmeter will be sent to the manufacturer to assess the calibration prior to conducting the performance test. A copy of the flowmeter calibration certificate will be attached to the test report.

We approve your use of the mass flowmeter in place of Method 2, 2A, 2C or 2D to measure the flare flow rate at the Richland Creek Road Landfill in Buford, Georgia. Since this alternative method is applicable to other similar facilities in this source category, we will be posting this letter on our website at http://www.epa.gov/ttn/emc/approalt.html for use by other interested parties.

If you have questions or would like to discuss the matter further, please call Foston Curtis at (919) 541-1063, or you may e-mail him at curtis.foston@epa.gov.

Sincerely,

Connie Oldham

Conniesue B. Oldham, Ph.D., Group Leader
Measurements Technology Group

cc: Foston Curtis (E143-02))
Karen Hays, Georgia EPD
David McNeal, Region 4

Date	Flare Temper	LFG Flow
	Deg F	SCFM
2/27/2026 0:02	902.14	1191.05
2/27/2026 0:12	902.41	1175.79
2/27/2026 0:22	836.07	1171.87
2/27/2026 0:32	865.52	1179.15
2/27/2026 0:42	886.12	1175.04
2/27/2026 0:52	881.4	1181.24
2/27/2026 1:02	870.1	1181.81
2/27/2026 1:12	901.81	1196.04
2/27/2026 1:22	893.7	1183.39
2/27/2026 1:32	878.08	1185.6
2/27/2026 1:42	920.35	1181.3
2/27/2026 1:52	864.65	1184.65
2/27/2026 2:02	927.33	1179.28
2/27/2026 2:12	930.19	1172.12
2/27/2026 2:22	916.77	1160.8
2/27/2026 2:32	911.05	1178.77
2/27/2026 2:42	905.66	1171.68
2/27/2026 2:52	908.79	1175.98
2/27/2026 3:02	899.48	1171.68
2/27/2026 3:12	924.14	1176.3
2/27/2026 3:22	901.68	1176.62
2/27/2026 3:32	918.83	1171.62
2/27/2026 3:42	898.49	1164.53
2/27/2026 3:52	898.75	1173.33
2/27/2026 4:02	878.81	1172.31
2/27/2026 4:12	840.46	1181.11
2/27/2026 4:22	879.67	1175.29
2/27/2026 4:32	870.17	1191.61
2/27/2026 4:42	862.46	1168.83
2/27/2026 4:52	862.99	1168.01
2/27/2026 5:02	863.85	1176.17
2/27/2026 5:12	810.08	1182.76
2/27/2026 5:22	874.82	1174.91
2/27/2026 5:32	835.27	1175.48
2/27/2026 5:42	859.27	1178.39
2/27/2026 5:52	818.32	1180.03
2/27/2026 6:02	839.33	1171.11
2/27/2026 6:12	852.75	1175.92
2/27/2026 6:22	839.79	1174.97
2/27/2026 6:32	842.72	1182.95
2/27/2026 6:42	837.07	1175.35
2/27/2026 6:52	865.85	1181.81

2/27/2026 7:02	885.13	1160.48
2/27/2026 7:12	838.93	1174.59
2/27/2026 7:22	862.59	1177.88
2/27/2026 7:32	864.12	1165.54
2/27/2026 7:42	860	1169.09
2/27/2026 7:52	868.97	1166.94
2/27/2026 8:02	852.82	1186.68
2/27/2026 8:12	874.09	1180.1
2/27/2026 8:22	874.22	1168.2
2/27/2026 8:32	879.14	1180.73
2/27/2026 8:42	854.22	1187.31
2/27/2026 8:52	853.82	1183.58
2/27/2026 9:02	868.18	1178.64
2/27/2026 9:12	875.95	1178.45
2/27/2026 9:22	884.13	1187.25
2/27/2026 9:32	872.36	1183.51
2/27/2026 9:42	857.94	1176.55
2/27/2026 9:52	875.29	1167.82
2/27/2026 10:02	776.18	1194.53
2/27/2026 10:12	813.2	1179.91
2/27/2026 10:22	845.38	1176.62
2/27/2026 10:32	738.23	1168.83
2/27/2026 10:42	771.39	1183.89
2/27/2026 10:52	809.81	1182.76
2/27/2026 11:02	784.02	1184.27
2/27/2026 11:12	759.36	1166.68
2/27/2026 11:22	805.69	1180.22
2/27/2026 11:32	745.01	1180.73
2/27/2026 11:42	736.5	1180.54
2/27/2026 11:52	834.01	1175.35
2/27/2026 12:02	815.6	1183.64
2/27/2026 12:12	816.26	1179.97
2/27/2026 12:22	792.07	1175.48
2/27/2026 12:32	805.03	1180.1
2/27/2026 12:42	709.38	1182.88
2/27/2026 12:52	854.35	1177.5
2/27/2026 13:02	768.74	1179.91
2/27/2026 13:12	644.24	1172.25
2/27/2026 13:22	655.54	1188.01
2/27/2026 13:32	826.1	1133.65
2/27/2026 13:42	799.91	1196.36
2/27/2026 13:52	791.93	1142.7
2/27/2026 14:02	702.33	1138.08
2/27/2026 14:12	770.66	1148.77

2/27/2026 14:22	790.21	1147.7
2/27/2026 14:32	897.69	1145.55
2/27/2026 14:42	842.78	1143.9
2/27/2026 14:52	734.1	1185.03
2/27/2026 15:02	730.91	1198.32
2/27/2026 15:12	683.32	1198.64
2/27/2026 15:22	811.68	1202.18
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2/27/2026 15:42	668.83	1198.13
2/27/2026 15:52	561.28	1202.44
2/27/2026 16:02	860	1194.02
2/27/2026 16:12	784.62	1197.44
2/27/2026 16:22	741.62	1185.48
2/27/2026 16:32	767.8	1209.97
2/27/2026 16:42	744.61	1205.28
2/27/2026 16:52	751.32	1211.99
2/27/2026 17:02	727.86	1206.55
2/27/2026 17:12	716.56	1205.35
2/27/2026 17:22	804.83	1210.16
2/27/2026 17:32	616.19	1200.66
2/27/2026 17:42	801.51	1213.32
2/27/2026 17:52	818.72	1202.63
2/27/2026 18:02	816.06	1205.85
2/27/2026 18:12	772.86	1209.52
2/27/2026 18:22	901.68	1202.69
2/27/2026 18:32	763.22	1194.46
2/27/2026 18:42	759.63	1209.59
2/27/2026 18:52	823.24	1205.16
2/27/2026 19:02	836.67	1195.16
2/27/2026 19:12	884.73	1212.94
2/27/2026 19:22	879.01	1201.61
2/27/2026 19:32	852.16	1191.49
2/27/2026 19:42	854.68	1210.03
2/27/2026 19:52	890.38	1213.64
2/27/2026 20:02	769.47	1201.68
2/27/2026 20:12	866.38	1203.13
2/27/2026 20:22	819.19	1197.82
2/27/2026 20:32	861.66	1202.31
2/27/2026 20:42	832.02	1198.45
2/27/2026 20:52	841.12	1198.83
2/27/2026 21:02	886.79	1194.53
2/27/2026 21:12	819.72	1195.29
2/27/2026 21:22	819.19	1189.4
2/27/2026 21:32	782.1	1190.6

2/27/2026 21:42	913.44	1186.05
2/27/2026 21:52	839.59	1196.23
2/27/2026 22:02	826.23	1193.64
2/27/2026 22:12	875.09	1187.38
2/27/2026 22:22	808.55	1194.08
2/27/2026 22:32	916.7	1176.87
2/27/2026 22:42	800.24	1175.86
2/27/2026 22:52	672.88	1185.35
2/27/2026 23:02	779.97	1178.33
2/27/2026 23:12	731.71	1180.79
2/27/2026 23:22	693.82	1183.26
2/27/2026 23:32	637.59	1164.15
2/27/2026 23:42	746.73	1183.58
2/27/2026 23:52	685.78	1179.15

APPENDIX B

FIELD AND CALCULATED DATA SHEETS

Wexford County Landfill
Manually Recorded Process Flow
27-Feb-26

Test 1			Test 2			Test 3			
Date/Time	Temp	Flow	Date/Time	Min	Max	Date/Time	Min	Max	
02/27/26 09:32	872.36	1183.51	02/27/26 10:12	813.2	1179.91	02/27/26 10:52	809.81	1182.76	
02/27/26 09:42	857.94	1176.55	02/27/26 10:22	845.38	1176.62	02/27/26 11:02	784.02	1184.27	
02/27/26 09:52	875.29	1167.82	02/27/26 10:32	738.23	1168.83	02/27/26 11:12	759.36	1166.68	
02/27/26 10:02	776.18	1194.53	02/27/26 10:42	771.39	1183.89	02/27/26 11:22	805.69	1180.22	
02/27/26 10:12	813.2	1179.91	02/27/26 10:52	809.81	1182.76	02/27/26 11:32	745.01	1180.73	
									Average:
		1180.5			1178.4			1178.9	1179.3

Net Heating Values

Test No.	Test C_i	K	HT
1	478000	3.32775E-05	15.91
2	475000	3.32775E-05	15.81
3	490000	3.32775E-05	16.31
	481000.00	Average:	**16.01**

Average Gas Quality

From Laboratory analysis received 3/16/26

o2	n2	ch4	co2
4.00	23.20	47.80	31.70
3.91	23.00	47.50	31.50
5.12	27.30	49.00	32.40
4.34	**24.50**	**48.10**	**31.87**

Stack Gas Exit Velocity

Test No.	Inlet (ft3/s)	Exhaust (ft2)	Exit Velocity (ft/s)
1	19.67	0.3491	56.36
2	19.64	0.3491	56.26
3	19.65	0.3491	56.29
Avg. Inlet Flow:	1179.27	Average:	**56.31**

Test No.	Avg Inlet Flow (cfm)
1	1,180.46
2	1,178.40
3	1,178.93
Avg.	1179.27

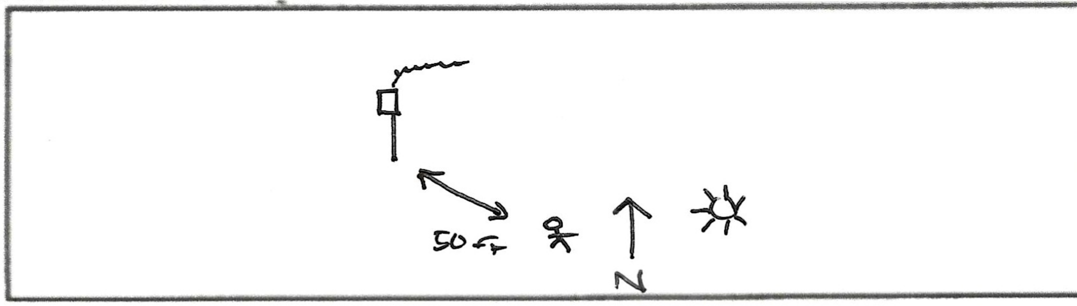
Log10 (Vmax)=(HT+28.8)/31.7

	Log10 (Vmax) =	1.410304259	
		1.407154968	
		1.42290142	
	Avg Ht =	1.413453549	
	Vmax (m/s)	(ft/s : m/s)	Vmax (ft/s)
	25.72197185	3.278688525	84.33
	25.53612341	3.278688525	83.72
	26.47899024	3.278688525	86.82
Avg Vmax =	25.90917288	3.278688525	84.95
			84.96

FUGITIVE OR SMOKE EMISSION INSPECTION
OUTDOOR LOCATION

Company Wexford County Landfill Location Manton, MI Company Rep. Anthony Testa	Observer T. Smith Affiliation EIL Date 02/27/26
Sky Conditions Partly Cloudy / Clear Precipitation None	Wind Direction N Wind Speed 10-20 mph
Industry Landfill	Process Unit open flare

Sketch process unit: indicate observer position relative to source; indicate potential emission points and/or actual emission points.



OBSERVATIONS

	Clock Time	Observation period duration, min:sec	Accumulated emission time, min:sec
Begin Observation	9:30	20:00	00:01
	9:55	20:00	00:01
	10:20	20:00	00:00
	10:45	20:00	00:01
	11:10	20:00	00:00
	11:35	20:00	00:00
End Observation	11:55	120:00	00:03

Figure 22-1

USEPA Method 3C/25C Field Data Sheet

Client: Wexford County Landfill   Date: 2/27/26
Source: open Flare   Run No.: 1

Tank No.: 50741   Tank Volume: 6L

Sample Tanks are leak - checked by the lab.
Confirm initial vacuum > 25" Hg (gauge): 20" "Hg

Sample Train Leak - Check

Allowable leak rate, $\Delta p = \frac{F \times P_b \times \phi}{V_t} \times 0.01$

F = Sample flow rate, cc/min
P_b = Barometric Pressure, cm Hg
ϕ = Leak - check period, min
V_t = Volume of train, cm^3

Volume of sample train = 19.31 cm^3
Volume of probe = 6.43 cm^3
V_t = 25.74 cm^3

$\Delta p = \frac{(100)(76)(5)}{(25.74)} \times 0.01$

Allowable Δp = 1.1 "Hg / 5 min.

Initial Vacuum: 20" "Hg
Final Vacuum: 20" "Hg
Actual Dp: 0" "Hg / 5 min.

Sample Train Data

Pre - test data			Post - test data		
P_b	28.35	"Hg	P_b	28.35	"Hg
T_{ti}	29	°F	T_{tf}	32	°F
P_{ti}	20"	"Hg	P_{tf}	2"	"Hg

Sampling Data

Clock Time	Elapsed Time	Train Vacuum	Rotometer (cc/min)	Comments
9:40	0	20"	100 cc	1168 cfm 1032 °F
9:45	5	17"		1180 cfm 919 °F
9:50	10	14"		1176 cfm 980 °F
9:55	15	11"		1170 cfm 997 °F
10:00	20	8"		1173 cfm 976 °F
10:05	25	5"		1163 cfm 1028 °F
10:10	30	2"		1173 cfm 1010 °F
	35			

Gas Flow to Flare (Pre-Sample Reading): Time: 9:32 1172 cfm 1005 °F
Gas Flow to Flare (Post-Sample Reading): Time: 10:13 1165 cfm 1010 °F

GEM meter reading through sample train
Time: 9:31

CH_4	45.0	%
CO_2	31.5	%
O_2	3.4	%
Balance	20.1	%

USEPA Method 3C/25C Field Data Sheet

Client: Wexford County Landfill Date: 2/27/26
Source: Open Flare Run No.: 2

Tank No.: 44640 Tank Volume: 6L

Sample Tanks are leak - checked by the lab.
Confirm initial vacuum > 25" Hg (gauge): 20 " "Hg

Sample Train Leak - Check

Allowable leak rate, $\Delta p = \frac{F \times P_b \times \phi}{V_t} \times 0.01$

F = Sample flow rate, cc/min
P_b = Barometric Pressure, cm Hg
ϕ = Leak - check period, min
V_t = Volume of train, cm^3

Volume of sample train = 19.31 cm^3
Volume of probe = 6.43 cm^3
V_t = 25.74 cm^3

$\Delta p = \frac{(100)(76)(5)}{(25.74)} \times 0.01$

Allowable Δp = 1.1 "Hg / 5 min.

Initial Vacuum: 20" "Hg
Final Vacuum: 20" "Hg
Actual Dp: 0" "Hg / 5 min.

Sample Train Data

Pre - test data			Post - test data		
P_b	28.35	"Hg	P_b	28.35	"Hg
T_{ti}	32°	°F	T_{tf}	33	°F
P_{ti}	20"	"Hg	P_{tf}	2"	"Hg

Sampling Data

Clock Time	Elapsed Time	Train Vacuum	Rotometer (cc/min)	Comments	
10:20	0	20"	≈ 100 cc/min	1168 cfm	995°F
10:25	5	17"		1171 cfm	978°F
10:30	10	14"		1167 cfm	967°F
10:35	15	11"		1170 cfm	976°F
10:40	20	8"		1167 cfm	998°F
10:45	25	5"		1181 cfm	998°F
10:50	30	2"		1169 cfm	1002°F
	35				

Gas Flow to Flare (Pre-Sample Reading): Time: 10:15 1180 cfm 1016 °F
Gas Flow to Flare (Post-Sample Reading): Time: 10:50 1168 cfm 1001 °F

GEM meter reading through sample train
Time: 10:14

CH4	46.2	%
CO2	32.0	%
O2	3.4	%
Balance	18.6	%

USEPA Method 3C/25C Field Data Sheet

Client: Wexford County Landfill
Date: 2/27/26
Source: Open Flare
Run No.: 3

Tank No.: 42721
Tank Volume: 6L

Sample Tanks are leak - checked by the lab.
Confirm initial vacuum > 25" Hg (gauge): 20 "Hg

Sample Train Leak - Check

Allowable leak rate, $\Delta p = \dfrac{F \times P_b \times \phi}{V_t} \times 0.01$

F = Sample flow rate, cc/min
P_b = Barometric Pressure, cm Hg
ϕ = Leak - check period, min
V_t = Volume of train, cm^3

Volume of sample train = 19.31 cm^3
Volume of probe = 6.43 cm^3
V_t = 25.74 cm^3

$\Delta p = \dfrac{(100)(76)(5)}{(25.74)} \times 0.01$

Allowable Δp = 1.1 "Hg / 5 min.

Initial Vacuum: 20" "Hg
Final Vacuum: 20" "Hg
Actual Dp: 0" "Hg / 5 min.

Sample Train Data

Pre - test data			Post - test data		
P_b	28.34	"Hg	P_b		"Hg
T_{ti}	33	°F	T_{tf}	35	°F
P_{ti}	20"	"Hg	P_{tf}	2"	"Hg

Sampling Data

Clock Time	Elapsed Time	Train Vacuum	Rotometer (cc/min)	Comments
11:00	0	20"	≈ 100 cc/min	1174 cfm 972°F
11:05	5	17"		1168 cfm 985°F
11:10	10	14"		1178 cfm 1024°F
11:15	15	11"		1180 cfm 988°F
11:20	20	8"		1178 cfm 1021°F
11:25	25	5"		1167 cfm 1003°F
11:30	30	2"		1160 cfm 968°F
	35			

Gas Flow to Flare (Pre-Sample Reading): Time: 10:53 1168 cfm
Gas Flow to Flare (Post-Sample Reading): Time: 11:30 1158 cfm

GEM meter reading through sample train
Time: 10:53

CH4 46.5 %
CO2 31.9 %
O2 3.2 %
Balance 18.3 %

USEPA Method 3C/25C Field Data Sheet

Client: Wexford County Landfill — Date: 2/27/26
Source: Open Flare — Run No.: 4

Tank No.: 34935 — Tank Volume: 6L

Sample Tanks are leak - checked by the lab.
Confirm initial vacuum > 25" Hg (gauge): 20 "Hg

Sample Train Leak - Check

Allowable leak rate, $\Delta p = \frac{F \times P_b \times \phi}{V_t} \times 0.01$

F = Sample flow rate, cc/min
P_b = Barometric Pressure, cm Hg
ϕ = Leak - check period, min
V_t = Volume of train, cm^3

Volume of sample train = 19.31 cm^3
Volume of probe = 6.43 cm^3
V_t = 25.74 cm^3

$\Delta p = \frac{(100)(76)(5)}{(25.74)} \times 0.01$

Allowable Δp = 1.1 "Hg / 5 min.

Initial Vacuum: 20" "Hg
Final Vacuum: 20" "Hg
Actual Dp: 0" "Hg / 5 min.

Sample Train Data

Pre - test data			Post - test data		
P_b	28.34	"Hg	P_b	28.35	"Hg
T_{ti}	35	°F	T_{tf}	38	°F
P_{ti}	20"	"Hg	P_{tf}	2"	"Hg

Sampling Data

Clock Time	Elapsed Time	Train Vacuum	Rotometer (cc/min)	Comments
11:40	0	20"	≈100 cc/min	1187 cfm 1006 °F
11:45	5	17"		1181 cfm 972 °F
11:50	10	14"		1171 cfm 967 °F
11:55	15	11"		1175 cfm 1017 °F
12:00	20	8"		1169 cfm 971 °F
12:05	25	5"		1180 cfm 1005 °F
12:10	30	2"		1164 cfm 877 °F
	35			

Gas Flow to Flare (Pre-Sample Reading): Time: 11:36 — 1167 cfm — 996 °F
Gas Flow to Flare (Post-Sample Reading): Time: 12:10 — 1164 cfm

GEM meter reading through sample train
Time: 11:32

CH4	45.8	%
CO2	31.6	%
O2	3.2	%
Balance	19.4	%

APPENDIX C

LABORATORY ANALYTICAL REPORT

1941 Reymet Road ● Richmond, Virginia 23237 ● Tel: (804)-358-8295 Fax: (804)-358-8297

Certificate of Analysis

Final Report

Laboratory Order ID 26C0175

Client Name:	EIL, LLC	Date Received:	March 2, 2026 10:14
	1323 Butterfield Road Suite 104	Date Issued:	March 16, 2026 17:53
	Downers Grove, IL 60515	Project Number:	260204
Submitted To:	Ben Kotrba	Purchase Order:	
Client Site I.D.:	Wexford Flare Performance Test		

Enclosed are the results of analyses for samples received by the laboratory on 03/02/2026 10:14. If you have any questions concerning this report, please feel free to contact the laboratory.

Sincerely,

Cedric Nelson

Group Leader

End Notes:

The test results listed in this report relate only to the samples submitted to the laboratory and as received by the Laboratory.

Unless otherwise noted, the test results for solid materials are calculated on a wet weight basis. Analyses for pH, dissolved oxygen, temperature, residual chlorine and sulfite that are performed in the laboratory do not meet NELAC requirements due to extremely short holding times. These analyses should be performed in the field. The results of field analyses performed by the Sampler included in the Certificate of Analysis are done so at the client's request and are not included in the laboratory's fields of certification nor have they been audited for adherence to a reference method or procedure.

The signature on the final report certifies that these results conform to all applicable NELAC standards unless otherwise specified. For a complete list of the Laboratory's NELAC certified parameters please contact customer service.

TNI Accredited
VELAP ID 460021

1941 Reymet Road ● Richmond, Virginia 23237 ● Tel: (804)-358-8295 Fax: (804)-358-8297

Certificate of Analysis

Final Report

Laboratory Order ID 26C0175

Client Name:	EIL, LLC	Date Received:	March 2, 2026 10:14
	1323 Butterfield Road Suite 104	Date Issued:	March 16, 2026 17:53
	Downers Grove, IL 60515	Project Number:	260204
Submitted To:	Ben Kotrba	Purchase Order:	
Client Site I.D.:	Wexford Flare Performance Test		

ANALYTICAL REPORT FOR SAMPLES

Sample ID	Laboratory ID	Matrix	Date Sampled	Date Received
W4	26C0175-01	Air	02/27/2026 12:10	03/02/2026 10:14
W3	26C0175-02	Air	02/27/2026 11:30	03/02/2026 10:14
W2	26C0175-03	Air	02/27/2026 10:50	03/02/2026 10:14
W1	26C0175-04	Air	02/27/2026 10:10	03/02/2026 10:14

ENTHALPY ANALYTICAL

1941 Reymet Road ● Richmond, Virginia 23237 ● Tel: (804)-358-8295 Fax: (804)-358-8297

Certificate of Analysis

Final Report

Laboratory Order ID 26C0175

Client Name:	EIL, LLC 1323 Butterfield Road Suite 104 Downers Grove, IL 60515	Date Received: Date Issued:	March 2, 2026 10:14 March 16, 2026 17:53
Submitted To:	Ben Kotrba	Project Number:	260204
Client Site I.D.:	Wexford Flare Performance Test	Purchase Order:	

ANALYTICAL RESULTS

Field Sample #: W4

Project Location:

Sample ID: 26C0175-01

Sample Matrix: Air

Sampled: 2/27/2026 12:10

Sample Type: LV

Canister ID: 063-00266::34935

Canister Size: 6L

Flow Controller ID:

Flow Controller Type: Passive

Initial Vacuum(in Hg): 21.2

Final Vacuum(in Hg): 2

Receipt Vacuum(in Hg): 4.4

Administrative Calc

Analyte	Results (%)	MDL (%)	LOQ (%)	Flag/Qual	Dilution	PF	Date/Time Analyzed	Analyst
Percent Dry	99.2	0.1	0.1		1	1	3/16/26 17:41	NSD

Volatile Organic Compounds by GC/TCD
ASTM D3588

Analyte	Results (BTU/ft³)	Flag/Qual	Dilution	PF	Date/Time Analyzed	Analyst
BTU Net or LHV	550		1	1	3/5/26 9:34	JTW
BTU Gross or HHV	611		1	1	3/5/26 9:34	JTW

Volatile Organic Compounds by GC/TCD - Unadjusted, as received basis
EPA 3C

Analyte	Result (Vol%)	MDL (Vol%)	LOQ (Vol%)	Flag/Qual	Dilution	PF	Date/Time Analyzed	Analyst
Methane, as received	47.3	0.45	0.45		9	1	3/4/26 16:38	JTW
Carbon dioxide, as received	31.3	0.45	0.45		9	1	3/4/26 16:38	JTW
Oxygen (O2), as received	4.12	0.45	0.45		9	1	3/4/26 16:38	JTW
Hydrogen (H2), as received	ND	0.18	0.18		9	1	3/4/26 16:38	JTW
Nitrogen (N2), as received	23.6	9.00	9.00		9	1	3/4/26 16:38	JTW
Carbon Monoxide, as received	ND	0.009	0.009		9	1	3/4/26 16:38	JTW

ENTHALPY ANALYTICAL

1941 Reymet Road ● Richmond, Virginia 23237 ● Tel: (804)-358-8295 Fax: (804)-358-8297

Certificate of Analysis

Final Report

Laboratory Order ID 26C0175

Client Name:	EIL, LLC 1323 Butterfield Road Suite 104 Downers Grove, IL 60515	Date Received: Date Issued:	March 2, 2026 10:14 March 16, 2026 17:53
Submitted To:	Ben Kotrba	Project Number:	260204
Client Site I.D.:	Wexford Flare Performance Test	Purchase Order:	

ANALYTICAL RESULTS

Field Sample #: W3

Project Location:

Sample ID: 26C0175-02

Sample Matrix: Air

Sampled: 2/27/2026 11:30

Sample Type: LV

Canister ID: 063-00061::42721

Canister Size: 6L

Flow Controller ID:

Flow Controller Type: Passive

Initial Vacuum(in Hg): 21.2

Final Vacuum(in Hg): 2

Receipt Vacuum(in Hg): 4

Administrative Calc

Analyte	Results (%)	MDL (%)	LOQ (%)	Flag/Qual	Dilution	PF	Date/Time Analyzed	Analyst
Percent Dry	99.3	0.1	0.1		1	1	3/16/26 17:41	NSD

Volatile Organic Compounds by GC/TCD
ASTM D3588

Analyte	Results (BTU/ft³)	Flag/Qual	Dilution	PF	Date/Time Analyzed	Analyst
BTU Net or LHV	393		1	1	3/5/26 9:34	JTW
BTU Gross or HHV	437		1	1	3/5/26 9:34	JTW

Volatile Organic Compounds by GC/TCD - Unadjusted, as received basis
EPA 3C

Analyte	Result (Vol%)	MDL (Vol%)	LOQ (Vol%)	Flag/Qual	Dilution	PF	Date/Time Analyzed	Analyst
Methane, as received	49.0	0.45	0.45		9	1	3/4/26 17:29	JTW
Carbon dioxide, as received	32.4	0.45	0.45		9	1	3/4/26 17:29	JTW
Oxygen (O2), as received	5.12	0.45	0.45		9	1	3/4/26 17:29	JTW
Hydrogen (H2), as received	ND	0.18	0.18		9	1	3/4/26 17:29	JTW
Nitrogen (N2), as received	27.3	9.00	9.00		9	1	3/4/26 17:29	JTW
Carbon Monoxide, as received	ND	0.009	0.009		9	1	3/4/26 17:29	JTW

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Final Report

Laboratory Order ID 26C0175

Client Name:	EIL, LLC 1323 Butterfield Road Suite 104 Downers Grove, IL 60515	Date Received: Date Issued:	March 2, 2026 10:14 March 16, 2026 17:53
Submitted To:	Ben Kotrba	Project Number:	260204
Client Site I.D.:	Wexford Flare Performance Test	Purchase Order:	

ANALYTICAL RESULTS

Field Sample #: W2

Project Location:

Sample ID: 26C0175-03

Sample Matrix: Air

Sampled: 2/27/2026 10:50

Sample Type: LV

Canister ID: 063-00415::44640

Canister Size: 6L

Flow Controller ID:

Flow Controller Type: Passive

Initial Vacuum(in Hg): 21.2

Final Vacuum(in Hg): 2

Receipt Vacuum(in Hg): 3.4

Administrative Calc

Analyte	% Results	% MDL	% LOQ	Flag/Qual	Dilution	PF	Date/Time Analyzed	Analyst
Percent Dry	99.4	0.1	0.1		1	1	3/16/26 17:41	NSD

Volatile Organic Compounds by GC/TCD
ASTM D3588

Analyte	BTU/ft³ Results	Flag/Qual	Dilution	PF	Date/Time Analyzed	Analyst
BTU Net or LHV	410		1	1	3/5/26 9:34	JTW
BTU Gross or HHV	455		1	1	3/5/26 9:34	JTW

Volatile Organic Compounds by GC/TCD - Unadjusted, as received basis
EPA 3C

Analyte	Vol% Result	Vol% MDL	Vol% LOQ	Flag/Qual	Dilution	PF	Date/Time Analyzed	Analyst
Methane, as received	47.5	0.45	0.45		9	1	3/4/26 18:20	JTW
Carbon dioxide, as received	31.5	0.45	0.45		9	1	3/4/26 18:20	JTW
Oxygen (O2), as received	3.91	0.45	0.45		9	1	3/4/26 18:20	JTW
Hydrogen (H2), as received	ND	0.18	0.18		9	1	3/4/26 18:20	JTW
Nitrogen (N2), as received	23.0	9.00	9.00		9	1	3/4/26 18:20	JTW
Carbon Monoxide, as received	ND	0.009	0.009		9	1	3/4/26 18:20	JTW

ENTHALPY ANALYTICAL

1941 Reymet Road ● Richmond, Virginia 23237 ● Tel: (804)-358-8295 Fax: (804)-358-8297

Certificate of Analysis

Final Report

Laboratory Order ID 26C0175

Client Name:	EIL, LLC 1323 Butterfield Road Suite 104 Downers Grove, IL 60515	Date Received: Date Issued:	March 2, 2026 10:14 March 16, 2026 17:53
Submitted To:	Ben Kotrba	Project Number:	260204
Client Site I.D.:	Wexford Flare Performance Test	Purchase Order:	

ANALYTICAL RESULTS

Field Sample #: W1

Project Location:

Sample ID: 26C0175-04

Sample Matrix: Air

Sampled: 2/27/2026 10:10

Sample Type: LV

Canister ID: 063-00994::50711

Canister Size: 6L

Flow Controller ID:

Flow Controller Type: Passive

Initial Vacuum(in Hg): 21.2

Final Vacuum(in Hg): 2

Receipt Vacuum(in Hg): 3.2

Administrative Calc

Analyte	% Results	% MDL	% LOQ	Flag/Qual	Dilution	PF	Date/Time Analyzed	Analyst
Percent Dry	99.4	0.1	0.1		1	1	3/16/26 17:41	NSD

Volatile Organic Compounds by GC/TCD
ASTM D3588

Analyte	BTU/ft³ Results	Flag/Qual	Dilution	PF	Date/Time Analyzed	Analyst
BTU Net or LHV	410		1	1	3/5/26 9:34	JTW
BTU Gross or HHV	456		1	1	3/5/26 9:34	JTW

Volatile Organic Compounds by GC/TCD - Unadjusted, as received basis
EPA 3C

Analyte	Vol% Result	Vol% MDL	Vol% LOQ	Flag/Qual	Dilution	PF	Date/Time Analyzed	Analyst
Methane, as received	47.8	0.45	0.45		9	1	3/4/26 19:11	JTW
Carbon dioxide, as received	31.7	0.45	0.45		9	1	3/4/26 19:11	JTW
Oxygen (O2), as received	4.00	0.45	0.45		9	1	3/4/26 19:11	JTW
Hydrogen (H2), as received	ND	0.18	0.18		9	1	3/4/26 19:11	JTW
Nitrogen (N2), as received	23.2	9.00	9.00		9	1	3/4/26 19:11	JTW
Carbon Monoxide, as received	ND	0.009	0.009		9	1	3/4/26 19:11	JTW

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Certificate of Analysis

Final Report

Laboratory Order ID 26C0175

Client Name:	EIL, LLC 1323 Butterfield Road Suite 104 Downers Grove, IL 60515	Date Received: Date Issued:	March 2, 2026 10:14 March 16, 2026 17:53
Submitted To:	Ben Kotrba	Project Number:	260204
Client Site I.D.:	Wexford Flare Performance Test	Purchase Order:	

Analytical Summary

Sample ID	Preparation Factors Initial / Final	Method	Batch ID	Sequence ID	Calibration ID
Volatile Organic Compounds by GC/TCD			**Preparation Method:**	**No Prep VOC GC Air**	
26C0175-01	1.00 mL / 1.00 mL	ASTM D3588	BJC0215	SJC0201	
26C0175-02	1.00 mL / 1.00 mL	ASTM D3588	BJC0215	SJC0201	
26C0175-03	1.00 mL / 1.00 mL	ASTM D3588	BJC0215	SJC0201	
26C0175-04	1.00 mL / 1.00 mL	ASTM D3588	BJC0215	SJC0201	

Sample ID	Preparation Factors Initial / Final	Method	Batch ID	Sequence ID	Calibration ID
Volatile Organic Compounds by GC/TCD - Unadjusted, as received basis			**Preparation Method:**	**No Prep VOC GC Air**	
26C0175-01	1.00 mL / 1.00 mL	EPA 3C	BJC0196	SJC0258	AJ50301
26C0175-02	1.00 mL / 1.00 mL	EPA 3C	BJC0196	SJC0258	AJ50301
26C0175-03	1.00 mL / 1.00 mL	EPA 3C	BJC0196	SJC0258	AJ50301
26C0175-04	1.00 mL / 1.00 mL	EPA 3C	BJC0196	SJC0258	AJ50301

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Certificate of Analysis

Final Report

Laboratory Order ID 26C0175

Client Name:	EIL, LLC 1323 Butterfield Road Suite 104 Downers Grove, IL 60515	Date Received: Date Issued:	March 2, 2026 10:14 March 16, 2026 17:53
Submitted To:	Ben Kotrba	Project Number:	260204
Client Site I.D.:	Wexford Flare Performance Test	Purchase Order:	

Volatile Organic Compounds by GC/TCD - Unadjusted, as received basis - Quality Control

Enthalpy Analytical

Analyte	Result	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit	Qual
Batch BJC0196 - No Prep VOC GC Air										
Blank (BJC0196-BLK1)					Prepared & Analyzed: 03/04/2026					
Methane	<	0.05	Vol%							
Carbon dioxide	<	0.05	Vol%							
Oxygen (O2)	<	0.05	Vol%							
Nitrogen (N2)	<	1.00	Vol%							
Hydrogen (H2)	<	0.02	Vol%							
Carbon Monoxide	<	0.001	Vol%							
LCS (BJC0196-BS1)					Prepared & Analyzed: 03/04/2026					
Methane	10200		ppmv	10100		101	80-120			
Carbon dioxide	9580		ppmv	10000		95.8	80-120			
Oxygen (O2)	1990		ppmv	2010		99.0	80-120			
Hydrogen (H2)	491		ppmv	500		98.2	80-120			
Nitrogen (N2)	4940		ppmv	5050		97.8	80-120			
Carbon Monoxide	516		ppmv	500		103	80-120			
LCS (BJC0196-BS2)					Prepared & Analyzed: 03/04/2026					
Methane	10200		ppmv	9880		103	80-120			
Carbon dioxide	9930		ppmv	9930		100	80-120			
Oxygen (O2)	5130		ppmv	4980		103	80-120			
Hydrogen (H2)	490		ppmv	498		98.4	80-120			
Nitrogen (N2)	5060		ppmv	4950		102	80-120			
Carbon Monoxide	522		ppmv	498		105	80-120			
Duplicate (BJC0196-DUP1)		**Source: 26C0207-03**			Prepared & Analyzed: 03/04/2026					
Methane	24.2	0.45	Vol%		24.6			1.74	5	
Carbon dioxide	49.7	0.45	Vol%		50.8			2.30	5	
Oxygen (O2)	<	0.45	Vol%		<0.45			NA	5	
Nitrogen (N2)	12.3	9.00	Vol%		12.5			1.63	5	
Hydrogen (H2)	10.1	0.18	Vol%		10.2			1.48	5	
Carbon Monoxide	0.13	0.009	Vol%		0.13			2.45	5	

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Client Site I.D.:	Wexford Flare Performance Test	Purchase Order:	

Volatile Organic Compounds by GC/TCD - Unadjusted, as received basis - Quality Control

Enthalpy Analytical

Analyte	Result	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit	Qual
Batch BJC0196 - No Prep VOC GC Air										
Duplicate (BJC0196-DUP2)		**Source: 26C0176-01**			Prepared & Analyzed: 03/04/2026					
Methane	22.7	0.45	Vol%		21.3			5.98	5	P
Carbon dioxide	56.5	0.45	Vol%		53.3			5.78	5	P
Oxygen (O2)	<	0.45	Vol%		<0.45			NA	5	
Hydrogen (H2)	20.3	0.18	Vol%		19.1			5.89	5	P
Nitrogen (N2)	<	9.00	Vol%		<9.00			NA	5	
Carbon Monoxide	0.01	0.009	Vol%		0.01			3.35	5	
Duplicate (BJC0196-DUP3)		**Source: 26C0176-02**			Prepared & Analyzed: 03/04/2026					
Methane	37.5	0.45	Vol%		37.6			0.160	5	
Carbon dioxide	40.7	0.45	Vol%		41.1			0.973	5	
Oxygen (O2)	0.58	0.45	Vol%		0.58			0.124	5	
Nitrogen (N2)	9.62	9.00	Vol%		9.62			0.0514	5	
Hydrogen (H2)	11.1	0.18	Vol%		11.1			0.346	5	
Carbon Monoxide	<	0.009	Vol%		<0.009			NA	5	
Duplicate (BJC0196-DUP4)		**Source: 26C0176-03**			Prepared & Analyzed: 03/04/2026					
Methane	33.4	0.45	Vol%		33.5			0.372	5	
Carbon dioxide	45.4	0.45	Vol%		45.6			0.543	5	
Oxygen (O2)	1.12	0.45	Vol%		1.12			0.0960	5	
Hydrogen (H2)	5.18	0.18	Vol%		5.19			0.274	5	
Nitrogen (N2)	17.0	9.00	Vol%		17.1			0.368	5	
Carbon Monoxide	<	0.009	Vol%		<0.009			NA	5	
Duplicate (BJC0196-DUP5)		**Source: 26C0176-04**			Prepared & Analyzed: 03/04/2026					
Methane	18.8	0.45	Vol%		19.0			0.756	5	
Carbon dioxide	46.4	0.45	Vol%		47.0			1.29	5	
Oxygen (O2)	<	0.45	Vol%		<0.45			NA	5	
Hydrogen (H2)	31.3	0.18	Vol%		31.5			0.686	5	
Nitrogen (N2)	<	9.00	Vol%		<9.00			NA	5	
Carbon Monoxide	<	0.009	Vol%		<0.009			NA	5	

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Submitted To:	Ben Kotrba	Project Number:	260204
Client Site I.D.:	Wexford Flare Performance Test	Purchase Order:	

Volatile Organic Compounds by GC/TCD - Unadjusted, as received basis - Quality Control

Enthalpy Analytical

Analyte	Result	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit	Qual
Batch BJC0196 - No Prep VOC GC Air										
Duplicate (BJC0196-DUP6)			**Source: 26C0175-01**		Prepared & Analyzed: 03/04/2026					
Methane	47.3	0.45	Vol%		47.3			0.0695	5	
Carbon dioxide	31.3	0.45	Vol%		31.3			0.162	5	
Oxygen (O2)	4.12	0.45	Vol%		4.12			0.0197	5	
Hydrogen (H2)	<	0.18	Vol%		<0.18			NA	5	
Nitrogen (N2)	23.6	9.00	Vol%		23.6			0.0212	5	
Carbon Monoxide	<	0.009	Vol%		<0.009			NA	5	
Duplicate (BJC0196-DUP7)			**Source: 26C0175-02**		Prepared & Analyzed: 03/04/2026					
Methane	49.0	0.45	Vol%		49.0			0.109	5	
Carbon dioxide	32.3	0.45	Vol%		32.4			0.0859	5	
Oxygen (O2)	5.11	0.45	Vol%		5.12			0.270	5	
Nitrogen (N2)	27.3	9.00	Vol%		27.3			0.0399	5	
Hydrogen (H2)	<	0.18	Vol%		<0.18			NA	5	
Carbon Monoxide	<	0.009	Vol%		<0.009			NA	5	
Duplicate (BJC0196-DUP8)			**Source: 26C0175-03**		Prepared & Analyzed: 03/04/2026					
Methane	47.4	0.45	Vol%		47.5			0.165	5	
Carbon dioxide	31.3	0.45	Vol%		31.5			0.590	5	
Oxygen (O2)	3.90	0.45	Vol%		3.91			0.192	5	
Nitrogen (N2)	22.9	9.00	Vol%		23.0			0.270	5	
Hydrogen (H2)	<	0.18	Vol%		<0.18			NA	5	
Carbon Monoxide	<	0.009	Vol%		<0.009			NA	5	
Duplicate (BJC0196-DUP9)			**Source: 26C0175-04**		Prepared & Analyzed: 03/04/2026					
Methane	47.7	0.45	Vol%		47.8			0.295	5	
Carbon dioxide	31.6	0.45	Vol%		31.7			0.375	5	
Oxygen (O2)	3.98	0.45	Vol%		4.00			0.331	5	
Hydrogen (H2)	<	0.18	Vol%		<0.18			NA	5	
Nitrogen (N2)	23.1	9.00	Vol%		23.2			0.195	5	
Carbon Monoxide	<	0.009	Vol%		<0.009			NA	5	

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Client Site I.D.:	Wexford Flare Performance Test	Purchase Order:	

Certified Analytes included in this Report

Analyte	Certifications	Analyte	Certifications
EPA 3C in Air			
Methane	VELAP		
Oxygen (O2)	VELAP		
Nitrogen (N2)	VELAP		

Code	Description	Laboratory ID	Expires
DURSC-NCDEQ	NCDEQ Durham Service Center	703	12/31/2026
DURSC-NCDHHS	NCDHHS Durham Service Center	37918	07/31/2026
MdDOE	Maryland DE Drinking Water	341	12/31/2026
NCDEQ	North Carolina DEQ	495	12/31/2026
NCDHHS	North Carolina Department of Health and Human	51714	07/31/2026
PADEP	NELAP-Pennsylvania Certificate #011	68-03503	10/31/2026
SCDES	South Carolina Dept of Environmental Services C	93016	06/14/2026
TXCEQ	Texas Comm on Environmental Quality #TX-C25-	T104704576	05/31/2026
VELAP	NELAP-Virginia Certificate #13761	460021	06/14/2026
WVDEP	West Virginia DEP Cert ID: WV-C25-00166	350	11/30/2026

Qualifiers and Definitions

P	Duplicate analysis does not meet the acceptance criteria for precision
RPD	Relative Percent Difference
Qual	Qualifers
-RE	Denotes sample was re-analyzed
PF	Preparation Factor
MDL	Method Detection Limit
LOQ	Limit of Quantitation
ppbv	parts per billion by volume
TIC	Tentatively Identified Compounds are compounds that are identified by comparing the analyte mass spectral pattern with the NIST spectral library. A TIC spectral match is reported when the pattern is at least 75% consistent with the published pattern. Compound concentrations are estimated and are calculated using an internal standard response factor of 1.

All EPA method 3C results are reported as normalized values when the sum total of all evaluated constituents is outside ± 10% of the absolute.

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AIR ANALYSIS
CHAIN OF CUSTODY

Equipment Due: 3/28/26

COMPANY NAME: EIL, LLC	INVOICE TO: Same	PROJECT NAME/Quote #: Wexford Flare Performance Test	
CONTACT: Ben Kotrba	INVOICE CONTACT:	SITE NAME: ~~Wexford Flare Sampling~~ Wexford County Landfill	
ADDRESS: 130 E Main St, Caledonia, MI 49316	INVOICE ADDRESS:	PROJECT NUMBER: 260204	
PHONE #: ~~646-558-8070~~ 984-415-3741	INVOICE PHONE #:	P.O. #:	
EMAIL: bkotrba@eilllc.com; tsmith@eilllc.com		Pretreatment Program:	
Is sample for compliance reporting? YES NO	Regulatory State: MI	Is sample from a chlorinated supply? YES NO	PWS I.D. #:
SAMPLER NAME (PRINT): Ben Kotrba	SAMPLER SIGNATURE: Benjamin Kotrba	Turn Around Time: Circle: 10 Days	

Matrix Codes: AA=Indoor/Ambient Air SG=Soil Gas LV=Landfill/Vent Gas OT=Other____________

063-26A-0030

Sampling Start Information — Barometric Pres. (In Hg): 28.35
Sampling Stop Information — Barometric Pres. (In Hg): 28.35

	CLIENT SAMPLE I.D.	Regulator Info: Flow Controller ID	Cal Flow (mL/min)	Canister Information: Canister ID	Size (L)	Cleaning Batch ID	LAB Outgoing Canister Vacuum (in Hg/psia)	LAB Receiving Canister Vacuum (in Hg)	Start Date	Start Time (24hr clock)	Initial Canister Vacuum (in Hg)	Starting Sample Temp °F	Stop Date	Stop Time (24hr clock)	Final Canister Vacuum (in Hg)	Ending Sample Temp °F	Matrix (See Codes)	ANALYSIS: 3C	BTU 3C	Percent Solids
1	W4	C-209	100	34935	6	BC260205	21.2"	4.4"	2/27/26	11:40	20"	35	2/27/26	12:10	2"	38	LV	x	x	x
2	W3			42721	6	BC260205	21.2"	4"		11:00	20"	33		11:30	2"	35	LV	x	x	x
3	W2			44640	6	BC260205	21.2"	3.4		10:20	20"	32		10:50	2"	33	LV	x	x	x
4	W1			50711	6	BC260210	21.2"	3.2		9:40	20"	29		10:10	2"	32	LV	x	x	x

RELINQUISHED:	DATE / TIME	RECEIVED:	DATE / TIME	QC Data Package	
Benjamin Kotrba	2/27/26 14:58	Fedex		Level I	☐
RELINQUISHED: Fedex	DATE / TIME	RECEIVED: [signature]	3.2.26 1014	Level II	☒
RELINQUISHED:	DATE / TIME	RECEIVED:	DATE / TIME	Level III	☐
				Level IV	☐

LAB USE ONLY

EIL, LLC 26C0175
Wexford County Landfill
Recd: 03/02/2026 Due: 03/16/2026

Therm ID: 310
Observed Temp °C: 18.0
Correction Factor °C: +0.1
Corrected Temp °C: 18.1
Ice: (Y N), Seal: (Y N)

v130325002

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Sample Conditions Checklist

Samples Received at:	**18.10°C**
How were samples received?	FedEx Express
Were Custody Seals used?	Yes
Are the custody papers filled out completely and correctly?	Yes
Do all bottle labels agree with custody papers?	Yes
Is the temperature blank or representative sample within acceptable limits or received on ice, and recently taken?	Yes
Are all samples within holding time for requested laboratory tests?	Yes
Is a sufficient amount of sample provided to perform the tests included?	Yes
Are all samples in appropriate containers for the analyses requested?	Yes
Were volatile organic containers received?	No
Are all volatile organic and TOX containers free of headspace?	NA
Is a trip blank provided for each VOC sample set? VOC sample sets include EPA8011, EPA504, EPA8260, EPA624, EPA8015 GRO, EPA8021, EPA524, and RSK-175.	NA
Are all samples received appropriately preserved? Metals (except Hg, B) do not require field preservation, but lab preservation may delay analysis. Field parameters performed by the lab are always received past holding time and will be noted as such.	Yes

APPENDIX D

SAMPLE CALCULATIONS

SAMPLE CALCULATIONS

Stack Gas Exit Velocity, V_s, feet per second (fps)

EPA method ALT-073 was utilized in place of Method 2C to determine the flow rate of the flare for use in calculating the exit velocity. This method allows for the process flow data from the mass flow meter in lieu of utilizing a Pitot tube to determine flow rate.

An average flow rate (cfm) for each test run was determined by averaging the process flow rates recorded during the duration of each test. These averages for each test run were then converted to ft^3/s. The example calculation for this conversion is below.

The average flow rate for Test 1 was 1180.46 cfm.

1180.46 cfm x 1 min/ 60 sec = 19.67 ft^3/s.

The flare exit tip is 8" inner diameter. The cross-sectional area of the flare tip is:

$A_s = \pi * (d/2)^2 / 144 = 3.14159 * (8/2)^2 / 144 = 0.3491\ ft^2$

The flare exit gas velocity for Test 1 is then:

19.67 ft^3/sec / 0.3491 ft^2 = 56.34 ft/sec

Maximum Permitted Velocity, V_{max}

The exit velocity calculated from the ALT-073 velocity data met the 60 fps limit of 63.11(b)(6)(i)(A), 60.18(c)(3)(i)(A) on all three (3) test runs, and as a three-test average. The general control devices requirements of 63.11(b)(6)(i)(A), 60.18(c)(3)(i)(A) also requires that the exit velocity be less than the calculated Maximum Permitted Velocity, V_{max} as determined by the equation presented in 63.11(b)(7)(iii), 60.18(f)(5):

$\text{Log10}\ (V_{max}) = (H_T + 28.8)/31.7$

The average net heating value, H_T, for the utility flare three test runs was 16.01 MJ/scm.

$\text{Log10}\ (V_{max}) = (16.01 + 28.8)/31.7 = 1.413$

$V_{max} = 10^{1.413} = 25.88$ meters per second.

There are 3.279 feet per meter, so V_{max}, in fps, is 3.279 ft/m * 25.88 m/s = 84.86 fps.

*Note: any slight differences between results present here, and elsewhere in the report, is due to rounding differences between spreadsheet and hand-held calculator.